5 Information and Communication Technologies in Education

Introduction

- All digital and non-digital devices, tools, content, resources, forums, and services that can be used for teaching and learning, expanding access to and the reach of resources, constructing capacities, and managing the educational system are included in the definition of 'information and communication technologies'(ICTs).
- These will also involve the creation of forums for interaction and exchange, as well as the development and deployment of platforms and processes for capacity development and content digitisation, deployment, and management.
- In addition to computers and software, these will consist of things like interactive digital content, internet and other satellite communication devices, radio and television services, online content repositories, online discussion forums, learning management systems, and management information systems.

Objectives of ICT in Education

- To blend technology with the conventional classroom teaching learning process.
- To broadcast educational lessons and provide online facilities for different subjects.
- To use electronic devices to develop awareness and psychomotor control.
- To carry out internet-based research to enhance the educational process.
- To encourage the students to use email, social networking sites, etc., for educational discussion, chats, etc.
- To facilitate video conferencing between teachers and students in the distance education programme.
- To exchange and share ideas among teachers for their professional growth.
- The development of ICT and ICT based tools has led to various important applications in the field of technology-based education like e-learning, smart learning, etc.
- In today's world, a classroom is different from the traditional one. ICT is changing methods of teaching and learning in educational institutions. The age of blackboards and chalk has now been replaced by smart boards and the internet.

Smart Diary

- Nowadays schools are going away from the troublesome plastic covered book embossed with the school's logo called a school diary or calendar, filled with teacher's remarks, or a note from parents to excuse the child from attending school. But now, cloud computing technology, such as Google Docs, Groups, Dropbox, etc., keeps students posted about events, projects, and homework. Children can no longer hide details about their homework as parents can simply log in and check on one of the portals with which the school has tied up.
- One of the major roles of technology in school life is real-time updates, for example, during heavy rains, accidents, or other disasters. For example, if a school bus is going to arrive late due to some reasons then parents can be informed through SMS. Schools are also installing GPS in their buses, which enables the school authorities to monitor the movement of their buses at any time.

Smart Board

- Schools have started using smart boards, which allow a teacher to do multitasking, such as project slides, show a video, or connect to the internet. Chalk, which teachers often used as a discipline tool by throwing little pieces at an inattentive child, will no longer be on the school stationery list. Schools are now having whiteboards and teachers are using brokers to write on them.

- If a teacher writes on a smartboard in one class, it gets replicated in another class. It also saves time. Teachers are now able to plan their lectures and load them on the board. The teaching process has become faster involving worksheets, classwork, or homework.
- Online assignments: Students seldom have to carry worksheets as even homework has gone online. Now, teachers create and post assignments online and share them with students. Neither teachers nor students have to worry about misplacing a file, everything is saved online, and accessible from anywhere. The system helps keep track of assignments given to students.
- Online forms and notices: Moreover, to go paperless, forms of any kind are on their way out. Instead of asking parents to fill out forms, schools are sending forms online. This is very helpful during admissions.
- Attendance: In a classroom of 50 to 60 students, about 10–15 minutes of 40 minutes are wasted in the teacher calling out students' names for attendance. The latest technology has been introduced, which can take care of students' attendance by face recognition technology within seconds.

Smart Classes

- The advancement of technology over the past several decades has brought about profound changes in every aspect of human existence. We still have teachers who use chalk and a chalkboard and textbooks and take the time to patiently explain complex ideas using just the resources at their disposal. Recently, though, rapid change has occurred. Everything in the classroom, from instruction to assessment, is being transformed by ICT. Complete classroom restructuring is now underway.
- What does it mean when we say that a class is 'smart?' It means that the class is using technology to assist instructors to adapt to changing classroom dynamics and improving their students' learning outcomes. Teachers are given easier access to relevant multimedia materials and the latest research findings that might enhance their lessons.
- The smart class places interactive technology next to the chalkboard, making it easier for professors to articulate their ideas and more certain that their students are grasping the material. Through the use of captivating images and animation, students can grasp challenging and complicated curriculum ideas. Consequently, kids' academic performance improves, and they have a better time studying. Educators can rapidly evaluate their pupils' progress and growth as learners.
- The instructor station in a smart classroom is used to operate the classroom's array of electronic tools (smart board, LCD projector, internet, etc.).
- The term 'smart class' simply refers to a modern and innovative approach to education. Smartboards may be mounted on a wall like a traditional blackboard, and a projector can be mounted on the ceiling to display content on the board. Getting educated is a breeze. The board functions similarly to a computer monitor and, if necessary, may be utilised in the traditional whiteboard fashion as well, making 'smart class' an efficient method of instruction. That is a radical departure from previous methods of teaching via word of mouth.
- Smart class is an advancement in today's technological landscape, and its use is destined to shake up the educational system.

e-Learning

- Another key creative use of computers in the learning process of the academic side of an educational institution is e-learning. e-learning, which may be CD-ROM based, network based, intranet based, or internet based, makes it possible to study from just about anywhere at almost any time, provided one has access to a well-equipped computer.
- Multimedia elements, such as text, music, video, animation, and 3D settings are all fair game. As a form of education, it has the potential to far outshine that of a traditional lecture hall. This kind of education allows students to study at their speed and get experience in the field.

- e-learning may fall prey to many of the same flaws as classroom training, including tedious presentations, repetitive voice, and a limited chance for interaction; the quality of electronic based training like other forms of training, lies in its content and delivery. But, the power of e-learning lies in the fact that modern software permits the development of highly efficient learning environments that can immerse the student completely in the subject matter.

Concepts of e-Learning

- 'Electronic learning' (e-learning) is a catch-all word for any kind of education that makes use of a computer. Nonetheless, it is often expanded to include the utilisation of portable technology like PDAs and MP5 players.
- Web based instructional materials and hypermedia in general, multimedia CD-ROMs or websites, discussion boards, collaborative software, email, blogs, wikis, text chat, computer assisted assessment, educational animation, simulations, games, learning management software, electronic voting systems, and more may all be a part of this.
- e-Learning may also refer to kid friendly educational websites that include things like printable worksheets and games. Businesspeople often use the word to refer to high-quality online education that does not break the bank. It has been spoken several times by various people. Some of them are given below.
    - The term 'e-learning' refers to the integration of online education.
    - e-Learning is a form of education that takes place in real-time on the World Wide Web.
    - e-Learning refers to the process of organising, delivering, selecting, managing, and extending education through the use of electronic networks.
    - For the purpose of education, e-learning harnesses the potential of networks, most prominently those that utilise internet technologies and, increasingly, satellite networks, and digital content.
    - Learners, content creators, and subject matter experts can all join a global online community through e-learning, which is defined as 'learning made possible through the use of the internet.'
    - e-Learning allows for more convenient and affordable education for more people and establishes a transparent system of responsibility for everyone involved in the educational process.
    - Businesses in today's dynamic environment that embrace e-learning provide their employees with the adaptability necessary to capitalise on new opportunities.

Multi-Tiered System of Online Learning

It may be broken down into four distinct tiers, from the most elementary to the most complex. Their names are

- Online support: Online support like knowledge databases, is a kind of e-learning. Forums, chat rooms, online message boards, email, and live instant messaging are all examples of online assistance options. Online help is somewhat more interactive than knowledge databases, allowing for more targeted queries and responses, as well as, faster resolution times.
- Knowledge databases: While not often considered training, knowledge databases constitute the foundation of e-learning. Knowledge bases are commonplace on software related websites, and they often include indexed explanations and help for software related issues, as well as, detailed how to guides for completing certain tasks. These tend to be highly interactive, letting you search the database with a keyword or phrase or choose from a list of options.
- Synchronous training: It is the one in which a real time teacher guides students through course material, known as synchronous training. At the appointed time, everyone signs in and has instantaneous access to the teacher and other students. A digital hand raiser and digital whiteboard are at your disposal. A session might run anything from a few hours to months or even years. Online platforms, such as websites, as well as, audio

and video conferencing, are often used for this sort of instruction. Telephony via the internet or even live, two-way broadcasts to a classroom full of pupils.

- Asynchronous training: E-Learning in its more conventional form is exemplified by asynchronous training. Learning at your speed with the help of a CD-ROM, network, intranet, or the internet is what this method entails. Communication with teachers is possible through email, online message boards, and chat rooms. There may not even be a need for a teacher, if the content stands on its own with supplementary connections.

Important Features of e-Learning

- Some salient characteristics of e-learning include the following, to summarise, online learning is
  - Designed around the learner, so students can choose content and tools that best suit their interests, needs, and skill levels; self-paced, so they can go at their own pace or take their time, if they need to; available around-the-clock, so they can fit it into their schedules; more effective for some learners than others because of the variety of delivery methods available; always available, so students can learn whenever it is convenient for them.
  - Because of the breakdown of borders, more people have access to education than ever before.
  - Student expenditures (tuition, housing, food, and childcare) are often decreased, and so is the amount of time spent travelling to and from classes. Students often benefit from more engagement with their peers and with their teachers, as well as, from improved computer and internet literacy.
  - Uses tried and true instructional strategies that have been refined throughout hundreds of years.
  - e-learning may contain a wide variety of features that make studying new information, a new procedure, or a new computer software more interesting and engaging. Making learning enjoyable or engaging is what boosts retention.
  - If you are not interested in what you are studying, you will not retain as much information. This is one of the numerous reasons why e-learning is so effective. Not every training course can be made available online. However, many people can and do so successfully.
- To ensure the success of e-learning, it is necessary to integrate the following elements.
  - Games, quizzes, and even the simple need to manipulate anything on the screen all generate greater curiosity, which in turn increases retention.
  - Images, sounds, and text all contribute to memory formation in different parts of the brain, leading to enhanced retention when used in combination.
  - Facilitating Communication between students and instructors in online courses e-learners may benefit greatly from the availability of real time communication tools like chat rooms, message boards, IM, and email, which can effectively substitute for in class discussions. The success of online programmes is heavily influenced by the quality of the online community that supports them.
  - Online courses have the unique ability to provide instantaneous responses to student questions and misconceptions. Since each stage of learning builds on the one before it, the quicker the feedback, the better. If there is no opportunity for correction, further development may be based on a misunderstanding.

Ease of Use and Portability of e-Learning

- It is not necessary to be there physically to participate in an online course.
- Your timetable is flexible, and you may access courses.
- Courses are accessible around the clock, giving you more freedom.
- Self-directed study (not too slow, not too fast).

- Your studies are not tied to any certain location; you may do them anywhere, even at your desk at work or while travelling.

Value and Buying Options

- Choose from a broad variety of courses; earn a degree, a certificate, or both.
- Return to school for a degree, to pick up a new trade, to hone your artistic abilities, or just for the experience.
- Anything you can imagine learning about, from painting to biology, can be done online, and at a cost that suits your budget.

Higher Retention

- When you study online, you will naturally gravitate toward what you find interesting.
- This, together with the fact that many various instructional strategies may be used to target students of varying learning styles, has been shown in studies to result in higher levels of retention than is often seen in a conventional classroom setting.

Flexibility

- You may tailor your online education to your interests and requirements since the emphasis is on you, the student. You can take classes taught by an instructor or study alone.
- Online education puts a world of educators at your fingertips. The technologies employed by online instructional designers enable the creation of resources that would be inaccessible in a conventional classroom setting.
- Students are better able to work together thanks to the availability of digital resources. Many projects require students to work together, and since online learning eliminates the need for students to be physically present with one another, it is a much more convenient (and pleasant) environment in which to do so.

Sequential Approach to ICT

- Research on the evolution of ICT in both developed and developing nations has uncovered at least four overarching strategies that educational systems and individual schools follow when adopting and using ICT. The figure below depicts a model of the continuum formed by the four strategies (emerging, applying, immersive/infusing, and transforming).
- The emerging approach: schools in the earliest phases of ICT development display the emerging approach. As a result, some of these institutions are beginning to acquire or receive donations of computer hardware and software. In this early phase, administrators and teachers are only beginning to examine the potential and repercussions of utilising ICT for school management and introducing ICT to the curriculum. Early-stage educational institutions are still heavily reliant on time-tested, teacher centred methods. There is an awareness of the benefits of ICT, and the curriculum emphasises a stronger foundation in the fundamentals. If an advanced strategy is desired, this curriculum can help facilitate that transition.
- The applying approach: this is most prevalent in institutions that have developed a fresh appreciation for the role of ICT in education. In this second stage, ICT is used by school administrators and teachers for routine tasks in administration and instruction. The classroom is largely controlled by teachers. When a school is at the 'applying approach' stage, it is making adjustments to the curriculum to incorporate more uses of ICT across the curriculum using targeted hardware and software. If an upgrade is desired, this curriculum can help facilitate that.
- The immersive/infusing approach: for schools that already use a variety of computer-based technologies in classrooms, laboratories, and administrative offices, the next step is the infusing approach, which entails integrating or embedding ICT across the curriculum. Educators investigate how the use of technology can enhance their work. The curriculum is starting to include cross-disciplinary approaches to better represent real-world contexts.
- The transforming approach: at the transforming approach are the schools that use ICT to reimagine and renew school organisation in innovative ways. 'Information and communications technology' (ICT) become

an unseen but essential component of our everyday lives. CT is taught as a distinct topic at the professional level and is integrated into all occupational fields, with a new emphasis on the learner and on the integration of subject areas in real world applications. Across the globe, schools have taken on the role of de facto community hubs for imparting knowledge.

ICT Enabled Teaching Learning Processes

- The term 'ICT enabled teaching learning' is used to describe a wide range of strategies for using information and communication technologies in the classroom. The instructor may use a wide range of ICT technologies for successful pedagogy, from supporting a course with media projection to multimedia self-learning modules to simulations to virtual learning environments. Each of these tools or strategies also necessitates some adjustment to the status quo in the classroom, if it is to be successful. The widespread availability of these types of educational resources is a key factor in accelerating the transition to ICT enabled classrooms.
- Schools having EDUSAT terminals, 'direct to home' (DTH) televisions, or other media equipment will include media-related events into their curriculum.
- Teachers will have a say in what kinds of digital information and tools are made available to their students. Additionally, they will be encouraged to create and distribute digital materials to their peers through digital repositories.
- There will be a transition from instructors using the computer lab for instruction and learning to the widespread use of ICT enabled classrooms.

Information and Communication Technologies in Schools

- Technology based and technology mediated tools for education have become more accessible because of advancements in information and communication networks. Because of this, ICT may now be used as a catchall system to supplement teaching and learning. Possible uses of ICT to address problems plaguing India's school system include.
- There is a pressing need to digitise and distribute instructional audio and video materials that are now dispersed over a wide variety of formats, languages, and media standards.
- Information and communication technologies may be used to effectively communicate data about and accelerate the process of adapting, adapting, translating, and disseminating scarce educational materials that exist in a wide variety of formats and media. This will facilitate its increased accessibility and wider adoption.
- The vastness of the school system and the reach of traditional means of training and support make it challenging to handle issues like teacher capacity development and continuous teacher support. However, ICT can help with both of these.
- Books, papers, handouts, charts, and posters have all found widespread usage in the classroom, but there is a dearth of print materials and online content in Indian languages. This is where information and communication technology (ICT) may be put to good use.
- Under utilising the potential and possibilities of computers and the internet by seeing them as just information delivery devices is a major waste. Educators and students alike might benefit greatly from the widespread availability of a wide range of apps, software tools, media, and interactive gadgets.

Computers and the Internet as a Tool for Open and Distance Education

- By using cutting edge ICTs, existing Open Schooling systems (including those at the national or state level) may be improved. The institutions will provide student support services, such as access to e-books, digital learning materials, Digital Repositories (with appropriate learning resources), etc. In addition, this will be put to use in expanding open and distant education for teachers.
- Students who have dropped out, are unable to complete their formal education, or are

enrolled in the non-formal education system have new options thanks to open and distance learning with the use of ICT. To better serve these students, open and distance learning systems will make use of ICT based education.

- Services including admissions, exams, e-accreditation, and grievance redressal would be given digitally, much like the National Institute of Open Schooling.
- These pupils will be included in the planned system to track students who have access to specialised educators. Online courses, online on-demand exams, digital repositories and content, media broadcasts planned through DTH/satellite based, open learning systems allowing multiple entries and exit points, opening out the school resources to non-formal students, guidance, and counselling, will result in the effective use of ICT for open and distance learning.

Helping Children with Disabilities Make Use of Information and Communication Technologies

- Screen readers, Braille printers, and other assistive technologies will be standard equipment in every classroom, and this accessibility focused ICT software will be embedded into the fabric of every school's network. Access to relevant information and communication technologies (ICTs) would be prioritised for students and instructors with special needs. With the help of ICT, we can finally realise the promise of education for all students.
- All educators will be made aware of the challenges faced by children with special needs and the opportunities presented by information and communication technologies. ICT enabled inclusive education will be a central tenet of all capacity building initiatives.
- Norms for accessibility will be implemented following the recommendations of the 'World Wide Web Consortium' (W3C) so that material may be viewed by children with special needs. Web based digital repositories that adhere to W3C standards will help fill the gap in accessibility for people with special needs. Talking books are only one example of the kinds of accessible digital material and services that will be created and distributed specifically for people with special needs.
- Accessibility standards shall be followed in the design of all programme related web based interfaces, including digital repositories, management information systems, etc.
- Lack of subject specific terminology in various Indian languages and unfamiliarity with cultural context might render digital communication and resources unavailable to students and instructors throughout the nation. The work of compiling suitable word lists and dictionaries in Indian languages will begin, and their broad translation will be actively supported.

School Management Information System (School MIS)

- There will be a national network of schools, educators, students, and administrators. The 'school management information system' (school MIS), a digital tool and content library, continuing education and professional development platforms, and guidance, counselling, and other student support services will all be part of this rollout.
- The MIS will be responsible for collecting data across a wide range of areas, including but not limited to: students, teachers, and their academic progress. Technology standards, such as language typefaces, word processors, technical dictionaries, etc., will likewise be defined by these norms. Universal access to data, media, and other resources will be protected by enforcing open standards.
- School MIS will become the central repository for all data pertaining to secondary education. This data will help researchers do their work more efficiently and educate policymakers at all levels of the educational system.

National Mission on Education through ICT (NMEICT)

- The 'Ministry of Human Resource Development' (MHRD) has launched the 'National Mission on Education via ICT' (NMEICT), a programme that aims to meet the requirements of students,

educators, and anyone else interested in learning by using technological advances.

- The Central Government of India has supported a programme called the National Mission on Education through ICT (NMEICT), which was officially started on February 3, 2009, in Tirupati, Andhra Pradesh.
- There is a significant gap in the levels of educational opportunities available in different parts of India. This initiative seeks to close the achievement gap by making high-quality learning materials and instructors available around-the-clock, seven days a week, for students of all backgrounds and socioeconomic statuses.
- Access, equality, and quality are the triumvirate of education policy ideals that must be met for the objective to be successful.
- All schools should have access to high-speed internet service. Affordability of internet and computers for schools. Creating excellent digital material.
- The primary goals of this mission are
    - To promote student agency and academic success via the use of IC to increase educational equality and expand access to the curriculum.
    - Over 400 schools and 22000 institutes of higher education throughout India would be linked. The goal is to raise the bar on teacher quality via the use of innovative synchronous training techniques.
    - Through the creative use of ICT, we can guarantee that all students, no matter where they live, have access to high-quality learning resources, even some that might otherwise be out of reach.
    - As a nation, we must ensure that none of our best minds go to waste.
    - All of Sakshat's services, including those offered via its content delivery site, should be provided without charge.
    - Making accessible e-content and e-educational films developed by leading educators in various fields, for use in both undergraduate and graduate programmes. As a result, the mission affords the country's educators and subject matter experts the chance to share their knowledge for the benefit of all students, helping to bridge the digital divide and provide education to those who may not otherwise have access.
    - The 'Indian Institutes of Technology' (IITs), the 'University Grants Commission' (UGC), the National Institutes of Technology (NITs), the 'Central Educational Commission', the 'International Institute of Distance Education' (IGNOU), and other prestigious educational institutions across the country are working with the Ministry of Human Resource Development (MHRD) to achieve the goals of this plan.
    - E-learning tools, virtual laboratories, open-source apps, etc., have benefited from the expertise of senior faculty members from a variety of academic institutions.
    - NMEICT has developed a new website to serve as a central hub for mission related information and to improve access to services, as well as, to encourage and collect user input on mission related initiatives.
    - To encourage students to continue their education throughout their lives, professors share their expertise for free on Sakshat. As such, it is anticipated that the NMEICT created material would mostly be distributed via this medium.

Policies of NMEICT

Broadband Connectivity to Colleges and Universities

- This mission aims to extend computer infrastructure and connectivity to over 26000+ colleges and 2000 polytechnics in the country including each department of 419 universities/deemed universities and institutions of national importance as a part of its motto to provide connectivity up to last mice.
- Up to 400 nodes, LAN on overage is also being provided under the mission. Till now, 400 universities and nearly 26000 colleges have been connected.

Low-Cost Access Device

- Even the best e-content cannot have a significant impact unless it reaches the vast majority of learners with ease when they demand it.
- Initially, the project of development of ultra-low-cost computing devices to enable students, wherever they may access education with entrusted to IIT Rajasthan. The main work was carried out on specification, design, tendering, and procurement of the first version of the Aakash tablet. The minister of Human Resource Development launched Akash 1 on 5 October, 2011.
- Subsequently, IIT Bombay took over the project related to testing, hardware, and software optimisation of low-cost access cum computing devices under the scheme of NMEICT.

e-Content Development Under NMEICT

- The various programmes under e-content development are
- NPTEL: it is a collaborative venture of IITs and IISC supported by this mission, and it offers e-learning through online web and video courses in the fields of engineering, science, and the humanities. By making high-quality engineering education resources available for free, NPTEL hopes to raise the bar throughout the nation.
- More than 329 courses have been developed and posted on the NPTEL website. More than 990 new courses in several fields of engineering and science are being developed as part of NPTEL's second phase.
- A 'consortium of educational communication' (CEC) has served as a test bed for the development of e-content for use in 'undergraduate' (UG) courses. e-Content for 19 UG topics was created by the CEC in phase I, while e-content for 68 UG subjects is being created by the CEC in phase II with the help of its media centres.
- Content generation of CEC includes content for BA History, BA English Language, BSc Botany, BSc Environmental Science, BA Mathematics, etc.
- Content generation of CEC is divided into four quadrants.

Second quadrant PDF/e-books/ illustration, video demonstration/ documents and Interactive simulations wherever required.	First quadrant (e-tutorial) Video and audio content in an organised form, animations, simulations, and virtual labs.
Third quadrant (web resources) related links, open content on the internet, case studies, historical development of the subject.	Fourth quadrant (self-assessments) MCQ, problems, quizzes, assignments and solutions, feedback through discussion, and clarifications on general misconceptions.

- e-Pathshala: MHRD, under NMEICT has assigned work to the UGC for the development of e-content in 77 subjects at the postgraduate level.
- The content and its quality is the key component of the education system. High-quality, curriculum based, interactive content in different subjects across disciplines of social sciences, arts, fine arts, and humanities, natural and mathematical, science is being developed under this initiative named e-PG pathshala.
- Cloud computing: We have already explained cloud computing. A common cloud computing infrastructure is developed by IIT Delhi, to offer multiple independent compute instances, in a scalable manner, based on individual project needs.
- Pilot development at NIC data centre with 640 compute cores and 50 TB + storage operational. 400 Computer cores have been tested in the second Development.
- NMEICT has launched virtual laboratories. Twelve different institutions are contributing to this initiative, which is being led by IIT Delhi.
- The overarching goal of this project is to create a virtual lab that can be used by as many students as possible to learn about different fields of

science and engineering. Virtual laboratories are primarily comprised of a user friendly graphical front-end, in sync with a backend, which may be a simulation engine operating on a server, or real measurement data.

- Experiments may be carried out at the user's location with no extra hardware or software setup needed when using the virtual lab. To conduct the studies remotely, all that is required is a computer terminal connected to a fast internet service provider. The 23rd of February, 2012, was the official national dedication of these laboratories.
- The goal of the 10-minute audio–video instruction on open-source software known as a 'spoken tutorial' is to educate students on crucial 'information technology' (IT) issues. These are made using the pedagogical (science of teaching) techniques established at IIT Bombay and are intended for individual study.
- If a youngster is struggling with English, they may watch these lessons with the spoken parts dubbed into any of the many Indian languages. With the right set of tutorials, even the most advanced information technology subject may be learned quickly and with little effort. Ultimately, we want to boost the student's employability.
- Over the previous three years, the spoken tutorial team at IIT Bombay has taught about four lakh students through self-workshops. Latex, PHP, Python, Linux, Jave, C, C++, etc., are all taught in these seminars.
- To phase out the usage of proprietary and commercial software in schools, the FOSSE initiative offers free assistance with 'free and open-source software' (FOSS). The scope of the project encompasses. Funding for a Personal FOSS Workshop.
- Lab Migration is an initiative to aid in the transition from closed-source software to free and open-source alternatives in academic research laboratories.
- Blended textbooks for student generated documentation of free/open-source software.
- e-Yantra (robotics in higher education): e-Yantra is an initiative to integrate robotics into engineering education to excite students by applying math, computer science, and engineering principles in exciting hands-on ways to produce engineers capable of supporting a dynamically expanding economy. e-Mission yantra is to improve the quality of engineering education in India by facilitating the teaching of embedded systems and robotics at the country's universities.
- ERP mission, with IIT Kanpur as the main institution in partnership with other partners in this project, aims to design, install, and administer a web-based software system for use by Indian academic institutes. Online admission system e-portfolio, general accounting system, payroll creation inside an institution, procurement and inventory management, library automation, etc., are only some of the major modules built in this system.
- To develop and provide structured education material to households in the most cost-effective way possible, the DTH Programme maintained 50 DTH educational channels 24 hours a day, 7 days a week, under the NME–ICT.
- It has the overarching purpose of closing the digital gap in higher education and ensuring that all students have 'access, dignity, and excellence.' Sixty-seven per cent of the population of India is potentially eligible to see the MHRD DTH programme. As part of this initiative, instructors will provide eight hours of live instruction per channel every day, with a second airing each day for students who were unable to tune in the first time around.

National Knowledge Network (NKN)

- The 'national knowledge network's' (NKN) overarching goal is to build a dependable and safe internal Indian network.
- With the help of NKN, institutions that are both forward-thinking and passionate will be able to access information and knowledge without regard to a physical location or the passage of time, reaping the advantages for themselves and society as a whole.

Features of NKN

- Facilitating teamwork in scientific inquiry and product advancement.

- Knowledge and data can be shared more easily, if high-speed network infrastructure is put in place.
- Connectivity throughout the NKN is achieved via gigabit-speed connections between the different universities. If India is ever going to become a knowledge society, the NKN is going to be an essential part of that infrastructure.
- NKN's architecture was created with high availability, high reliability, and scalability in mind. The network is built around a blazing fast core with speeds ranging from multiple 2.5/10 G to 40/100 Gbps. A distribution layer is added on top of the core to increase its functionality.
- In addition, NKN facilitates the development of VPNs for sub communities. Users benefit from its global reach and cutting-edge features.
- Promoting a very fast infrastructure for electronic government.
- Making it easier to pursue graduate level studies in a technical or scientific discipline without leaving home.
- Increasing the efficiency with which various networks in the academic, medical, business, and governmental spheres may collaborate.

Philosophies of NKN Design

- To serve as the backbone of the network, analogous to a national highway, on which all users may travel.
- The goals of this project are to construct a network that is capable of growing in both size (across the nation) and throughput. In addition, to encourage, enable, enhance, and empower the user is the core belief.

Practical Applications of NKN

- In the NKN's virtual classroom, instructors and students may engage in two-way communication in real time, allowing for more productive distance education. Lectures, presentations, and classroom handouts may now be shared across universities through the network.
- Virtual library: the virtual library is an application of NKN that facilitates the exchange of scholarly materials, such as journals, books, and research papers across universities.
- Government data centres and networks are only two examples of the kinds of e-governance infrastructures that the NKN can help to connect. The transmission of large amounts of data are a need for e-governance applications, and NKN offers this service.
- The NKN makes it possible for researchers from various institutions to work together on studies. Access to state-of-the-art research infrastructure and the exchange of scientific data are both made possible by NKN.
- High-performance computing is vital for national security, industrial productivity, and scientific and technological advancements; hence, these resources must be shared. Networking provides widespread access to high-performance computing, allowing for cutting edge research in computationally demanding disciplines including meteorology, seismology, engineering, and others.

Education Satellite (EDUSAT)

- On 20 September, 2004, India's educational programme received a boost with the launch of 'Education Satellite' (EDUSAT). EDUSAT is a satellite developed locally that is used for educational purposes only. This represents a breakthrough in the evolution of the idea of 'the education.' EDUSAT now resides in a geostationary orbit 36000 km above the Earth's surface.
- The first satellite designed specifically to support the educational community, EDUSAT has made history. The system is tailored to accommodate the rising need for a high-quality, interactive, satellite-based distance education system across the country's major urban centres.
- The EDUSAT, also known as GSAT 3, is the first satellite dedicated solely to supporting the educational sector. It is optimised for use with a digital interactive classroom and multimedia multicentric system in an AV setting.
- First successfully demonstrated in India in 1975–76, the Satellite Instructional Television

Experiment proved that educational programming could be successfully beamed via satellite (SITE). About 2400 villages in India across six states received direct broadcasts of health, hygiene, and family planning programmes as part of this ground-breaking experiment, which has been called the largest sociological experiment ever conducted.

- The satellite has many regional beams, each of which can be pointed at a different part of India. The primary goal of EDUSAT is to improve access to formal education at all levels (primary through graduate), but it will also facilitate non formal education and developmental communication.
- Students living in remote or economically depressed areas can benefit greatly from tele education because it allows them to participate in their education without having to incur the high costs associated with travelling to urban centres. The primary focus of this initiative is to improve educational opportunities for students in underserved communities.
- It is evident that there is a great need to provide such support for the betterment of rural education in India, and that there are very few teaching experts available in rural areas. The benefits are many, including the fact that the financially disadvantaged student does not have to spend time and money travelling to metropolitan study centres.
- In each case of remote 'satellite interactive terminal' (SIT), the instructor in the studio may have a two-way conversation. Both the Satellite Interactive Terminal (SIT) and the 'receive only terminal' (ROT) are part of this system (ROT).

Education and Research Network (ERNET)

- 'Education and Research Network' (ERNET) India is an autonomous scientific society under the administrative control of the Ministry of Information Technology, Government of India.
- ERNET has made a significant contribution to the emergence of networking in the country. The focus of ERNET India is not limited to just providing connectivity, but to meeting the entire needs of the academic and research institutions by providing constancy, project management, training, and other value-added services, such as web hosting, email services, video conferencing, domain registration, CUG services.
- ERNET India is serving institutions in various sectors, namely, health, agriculture, higher education, schools, and science and technology.
- It also supports for campus area network and currently serves over 1300 institutions.

e-Basta

Using e-Basta, approved educators at any given institution may curate individualised sets of course materials, textbooks, and electronic books for their students. The National Council for Educational Research and Training (NCERT) has made accessible in Hindi, English, Urdu, and Sanskrit 329 electronic textbooks for students in grades 1 through 12. Increasingly, academic institutions are allowing students to access course materials online and save them on their mobile devices or personal computers.

Usage of the e-Basta Mobile Application

- The e-Basta App gives students access to the e-Basta. The e-Basta app, it is formatted in a way that is simple for students to use. As long as the app has access to the correct e-Basta structure, it will produce the material according to the teacher's or school's specifications, regardless of the actual file names or locations.
- The e-Basta site would serve as a hub for the organisation's three main constituencies—students, educational institutions, and content providers.
- Educational Institutions: Institutions of higher learning may search the library's holdings, then choose and assemble the necessary course materials in electronic format for their students' use. This allows them to set up an e-Basta for each of their classes and provide their students easy access to the digital materials they need.
- By uploading the e-content of their study materials, publishers may make them available

to schools and students via e-Basta. It would be up to the school and the instructors to pick, which books and chapters pupils may read. All of these options would be accessible from a centralised dashboard that teachers and school administrators could access.

- To access the e-Basta portal and its collection of curricular resources, students need to get the e-Basta app for their computers or mobile devices. Both the institution and the book publishers would have previously decided which books would be available for download.

Virtual Classroom (VCR)

- A 'virtual classroom' (VCR) is an enhanced learning environment, constructed utilising the internet, computers, and supplicated video conferencing technologies, in which either instructor is not physically present (for remote learning) or pupils are not present (distance education) in the classroom.
- The purpose of VCR is
  - To aggregate academic resources hence, providing access to advanced educational experiences.
  - To provide real time, online instruction for distant students.
  - This is an effort to make classroom instruction more efficient and successful by bringing together students and teachers.
  - To hold and participate in the meetings, webinars, conferences/symposium/ workshop, interviews, etc., using video conferencing.

Advantages of Virtual Classroom

- Ensures comprehension: Ensure students comprehend their teachings by asking for rapid feedback, addressing questions, and delivering in-depth verbal explanations of a complicated subject.
- Offers anytime access: Offer live courses, office hours, and group discussions at times that are convenient for teachers and students, not only when the physical facilities are open.
- Recording of the class: It would be amazing, if you could study for a test by travelling back in time to the original class. All courses may now be seen in their entirety since recordings of the live event are kept indefinitely, including all audio, chat, and material.
- Electronic whiteboard with polls, quizzes, and surveys: An excellent practise of live training is to routinely present interactive activities. By presenting whiteboard tasks and asking polling questions, kids will stay very engaged.
- Multiple learning styles: enable children to fulfil their developmental goals by appealing to different learning styles since many pupils are auditory and visual learners. Bring together kids and teachers who would not otherwise talk to one other and foster a feeling of belonging in the classroom.
- Students with disabilities should have the same opportunities to participate in live online instruction as any other student. Accessible features include closed captioning for the hearing impaired, keyboard shortcuts and hotkeys for the visually impaired, and support for the majority of screen readers. Thanks to these accessibility enhancements, this technology will be available to everyone.
- The more reserved student might feel more at ease joining in discussions with their more outgoing peers thanks to our public and private text chat features.
- With the help of application sharing, you and your students may work together on projects or learn about complex software, as if you were in the same room. Anyone else online will be able to see what you are doing in whatever programme you are using thanks to application sharing. It is possible to hand over the reins so that pupils may work in tandem with you.
- Digital classroom brings together cutting-edge tools with tried and true pedagogical methods to provide an optimal learning environment for students. Teachers may get some assistance with digital lesson planning and delivery via the Digital classroom.
- More than 800 multimedia instructional materials (for IITs, Polytechnics, and Engineering institutions) and presentation tools and technologies are provided by digital classrooms.

Massive Open Online Course (MOOC)

Massive open online courses (MOOCs) are courses designed for large numbers of participants that can be accessed by anyone anywhere as long as they have an internet connection, are open to everyone without entry qualifications, and offer a full/complete course experience online for free.

- M: massive: an online course designed for a large number of participants.
- O: open: course can be accessed by (almost) anyone anywhere as long as they have an internet connection.
- O: online: complete course online.
- C: course: the course offers a full course experience including facilitation interaction among peers (including some but limited interaction with academic staff).

Study Webs of Active Learning for Young Aspiring Minds (SWAYAM)

- 'Study Webs of Active Learning for Young Aspiring Minds' (SWAYAM's) overarching goal is to provide a unified framework for reshaping the way children learn across all grade levels and subject areas.
- The Ministry of Human Resource Development (MHRD) of the Indian government conceived of this programme after extensive consultation with all relevant parties, including educators at both Central and State institutions, students, and testing organisations. It will be a "Mission Mode Project (MMP)" that follows the 'National e-Governance Plan's standards' (NeGP).
- With an increased emphasis on self-learning initiatives as a key activity for our aspirational youths to pursue the right career choice or learning capabilities so that they join the workforce with better skills to enhance growth and development, the Ministry of Human Resource Development (MHRD) envisions setting up a comprehensive MOOCs Platform to enable course takers to enrol in a course online and at no cost.
- Additional chances will be made available to students from farther afield thanks to the reach and accessibility of online education. There will likely be around 2000 courses available on SWAYAM, covering a wide range of subject areas.

Digital Saksharta Abhiyan (DISHA)/National Digital Literacy Mission (NDLM)

- Digital Literacy is the ability of individuals and communities to understand and use digital technologies for meaningful actions within life situations.
- The 'National Digital Literacy Mission' (NDLM) has come a long way since its inception in August 2012. The programme started as an industry initiative with nine partners in the IT industry coming together to accelerate digital literacy in India in keeping with the government's vision of having one e-literate person per household.
- The goal is to help people acquire the foundational ICT skills they need to utilise IT and associated applications, become more engaged in the political process, and improve their economic prospects.
- One member of every eligible family in strategically chosen blocks throughout all States and Union Territories will get ICT training as part of the National Digital Literacy Mission's (NDLMs) first plan.
- Individuals should have access to information, knowledge, and skill through digital technology.

The Indian Digital Library

- The 'Digital Library Initiatives' (DLI) original goal was to create a searchable and browse able digital library of books. The books are digitised at many sites throughout the world. High performance, scalability, availability, and cost-effectiveness were all taken into account when designing the DLI project's underlying architecture.
- To ensure that all major works of literature, art, and science are preserved for future generations and may be accessed by anyone, anywhere in the world, at any time, the 'Digital Library of India' (DLI) initiative was created.
- Presently, the project is focused on the digitisation and preservation of books; however, preserving pre-existing digital

assets of other types (video, audio, etc.), is a potential future direction. Different locations in India are responsible for scanning and archiving digital material.

Practice Exercise

1. Consider the following as advantages of ICT tools in educational systems.

I. Increased capacity and cost-effectiveness of the educational system.
II. Achievement by target groups that had limited access to traditional education.
III. Support for improvement of the quality and relevance of existing structures of education.
IV. Provision of links between various educational institutions for knowledge sharing.

Which of the above statements are correct?

(A) I, II, and III only
(B) I, II, and IV only
(C) III and IV only
(D) I, II, III, and IV

2. Consider the following objectives regarding ICT tools in technology-based learning.

I. Expanding access to all levels of education.
II. Improving the quality of education.
III. Enhancing lifelong learning.
IV. Facilitating non formal education.

Which of the following is true?

(A) I, II, and III
(B) I and II
(C) I, II, and IV
(D) I, II, III, and IV

3. Consider the following statements regarding e-Basta.

I. For students they provide easy access to interactive and dynamic content augmented with text, charts, graphics, videos, and auxiliary resource.
II. e-Bastas are delivered through the internet with no packaging no shipping and hence, no wait.
III. The e-Books and other contents in the e-Basta can be logically organised by the school/teacher while creating the e-Basta. Teachers can choose and bundle content according to their teaching methods.
IV. For publishers it gives convenience to register, sell and distribute e-books and related interactive e-contents online.

Which of the following is true?

(A) I, II, and III
(B) I and II
(C) I, II, and IV
(D) I, II, III, and IV

4. Consider the following statements about e-learning.

I. It provides platform independent delivery and seamless integration of multimedia content.
II. It can be categorised into synchronous and asynchronous learning.
III. Web based learning is part of e-learning.
IV. It is internet enabled learning.

Which of the following is true?

(A) I, II, and III
(B) II, III, and IV
(C) I, II, and IV
(D) I, II, III, and IV

5. Assertion (A): Digital libraries/online libraries provide improved access, improved preservation, and scalability.

Reason (R): Digitisation is a process of taking library material in the form of books and converting them into an electronic form where they can be easily manipulated.

(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

6. Which of the following are the features of educational technology?

I. e-learning
II. Instrumental technology
III. ICT in education
IV. Multimedia learning
V. Computer aided education
VI. Abacus bared education

(A) I, II, III, IV, and V
(B) I, II, III, and IV
(C) I, II, and III
(D) All of the above

7. Which amongst the following statements is/are correct?

I. Learner group in e-learning is more homogeneous while in classroom learning more heterogeneous.
II. Information and Communication Technology (ICT) based education can bring a better link between theory and practice.

Select the correct code.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

8. Information and communication technology (ICT) based education can never completely replace classroom education, the most important reason is:

(A) Hardware and software in ICT change frequently.
(B) Classroom education is existing for long.
(C) Face to face communication is most enriched.
(D) e-Learning needs e-content including e-books.

9. Which amongst the following statements is/are correct about National Mission on Education through ICT (NMEICT)?

I. It is under the Ministry of Human Resources Development of the Government of India and focuses on the primary education.
II. Its focus areas include interactive multimedia content, high-speed connectivity, low-cost access devices, and direct to home (DTH) education channels.
III. Sakshat is a one-stop education portal.

Select the correct code.
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III

10. Which amongst the following statement is incorrect about Education and Research Network (ERNET)?

I. It is under the Ministry of Human Resources Development.
II. It provides support for IPv6, domain, registration, and email services to colleges and universities.
III. It gives support for high-speed campus area network (CAN).

Select the correct code.
(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

11. 'SWAYAM', an initiative of the Government of India, aims at:

(A) Promoting the Self-Help Groups in rural areas.
(B) Providing financial and technical assistance to young start-up entrepreneurs.
(C) Promoting the education and health of adolescent girls.
(D) Providing affordable and quality education to the citizens for free.

12. Consider the following statements regarding using ICT tools in education.

I. It helps students to learn from anywhere.
II. It provides round-the-clock service.
III. It provides scalability.
IV. It helps the student to get a quality education in remote places.

Select the correct answer using the codes given below.
(A) I and II
(B) I, II, and III only
(C) I, II, and IV only
(D) I, II, III, and IV

13. Consider the following of ICT in technology-based learning.

I. Helps in diverse learning
II. Provides convenience and portability
III. Provides global collaboration
IV. Provides flexibility

Select the correct code.

(A) II, III, and IV
(B) I, II, and III
(C) I, II, and IV
(D) All of the above

14. Consider the following statements regarding ICT, which of the following is correct?

(A) It helps in process simplification.
(B) It expands education to the workplace.
(C) It enables easy, convenient sharing.
(D) It includes hardware, and software connected to the computer only.

15. National Mission on Education through ICT (NMEICT):

(A) Is a mission to promote quality education through multimedia e-content like e-books and e-journals
(B) Provides distance education through DTH
(C) Connects universities and colleges
(D) All of the above

16. e-Basta is:

(A) School books to e-Books
(B) A method to school books on learning tablets
(C) Both (A) and (B)
(D) None of the above

17. Consider the following statement about education and research network (ERNET).

I. It gives support to IPV6.
II. It provides support for a high-speed campus area network.

Which of the following is false?

(A) Both I and II
(B) I only
(C) II only
(D) None of these

18. Which of the following is a false statement about e-learning?

(A) It is internet enabled learning.
(B) Assistive technologies like speaking tables help disabled persons.
(C) It helps in making a paperless environment and provides scalability in the system.
(D) It is independent of the service provider.

Answer Key

1. (D)	**2.** (D)	**3.** (D)	**4.** (D)	**5.** (A)	**6.** (D)	**7.** (B)	**8.** (C)	**9.** (C)	**10.** (B)
11. (D)	**12.** (D)	**13.** (D)	**14.** (D)	**15.** (D)	**16.** (C)	**17.** (D)	**18.** (D)		